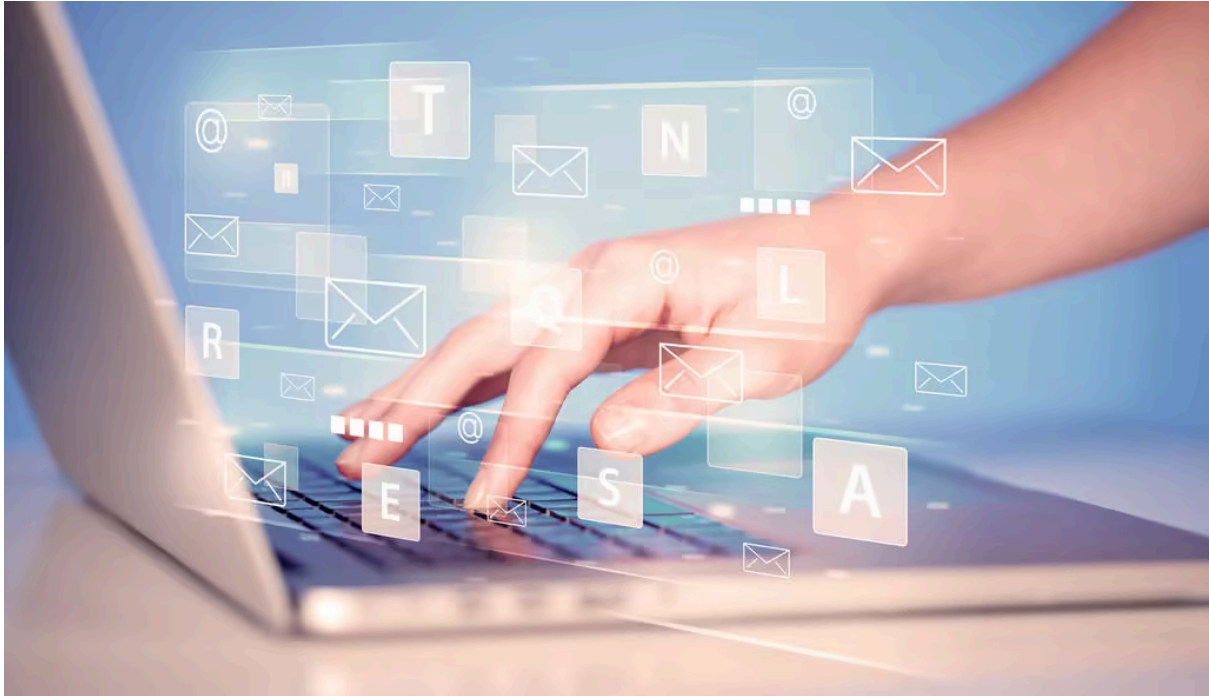


Phase 2: Advanced testing techniques

2.1 Test Automation:

- understanding the benefits and challenges of test automation?



Ans:- What is Test Automation:

Automation testing is the process of using software scripts and tools to execute and validate tests automatically. It involves using automation software to control the implementation of tests, compare actual results with anticipated results, and report the results of the tests.

Test automation can be used for a series of testing activities, counting [regression testing](#), [functional testing](#), performance testing, and load testing. It is generally used in software development to enhance the accuracy and efficiency of testing, as well as to reduce the cost and time of testing. Test automation can be done using a series of frameworks and tools, including commercial and open-source tools.

Why is Test Automation important?

Test automation is significant for several reasons:

1. **Enhanced Efficiency:** Test automation can save a huge amount of time and effort by automating monotonous tasks and freeing up QA testers to focus on complicated and critical testing deeds.
2. **Quality Assurance:** Test automation in software engineering can assist to ensure that software products meet essential quality standards by providing accurate and fast feedback on issues and defects, allowing test automation development teams to address them efficiently and rapidly.
3. **Improved Scalability** – Automating your tests transforms the scale at which your test QA team operates as PCs can run tests 24 hours a day. This lets you run a lot more tests with similar resources.
4. **Increased Test Coverage:** Automated tests can execute many test cases rapidly and accurately, allowing QA testers to cover an extensive range of scenarios and edge cases that might be hard to test manually.
5. **Reduced Time-to-Market:** By automating testing, development teams can accelerate their launch/ release cycles and rapidly deliver top-quality products to market.
6. **Consistency and Cost-effective:** Automated tests are implemented the same way every time, ensuring consistency. It can also be a lucrative way to test software products, as it lessens the related costs of maintaining a large testing team.

Why is Automation Testing core to DevOps?

Automation testing is core to DevOps for multiple reasons:

1. **Continuous Testing:** In DevOps, testing is incorporated into every stage of the [SDLC](#) (software development lifecycle), from code development to deployment. Automated testing allows QA teams to test iteratively and constantly throughout the test automation development process, guaranteeing that software is of top quality at every stage.
2. **Collaboration:** DevOps is all about collaboration between operations, development, and testing teams. Automated testing

frameworks and tools can help enable this teamwork by providing a common language and series of tools that all teams can use.

3. **Scalability:** As DevOps teams scale and the complexity of the software being developed rises, automated testing becomes vital to ensure that software is comprehensively tested and of high quality.
4. **Continuous Delivery:** In DevOps, the goal is to deliver software reliably and quickly. Automated testing is essential to achieving this objective by ensuring that software is thoroughly tested before it is released, minimising the risk of defects and issues in production.
5. **Detect bugs and Save time-Unlike manual testing.** Automatic test cases easily detect bugs and make testing faster which means you can deploy products to market more rapidly.
6. **Security:** Move quickly without compromising compliance and security by leveraging automated compliance policies, configuration management techniques, and fine-grained controls.

Benefits and Limitations of Test Automation

Test automation proffers numerous benefits and limitations, which are discussed below:

Benefits of Test Automation:

1. **Speedy Tests:** Automated testing is much quicker than manual testing, letting QA experts or testers test more scenarios in a shorter time.
2. **Increased Test Coverage:** Automated testing can cover an immense number of test cases and scenarios, counting edge cases and negative tests that are tough to manually test.
3. **Consistency:** Automated testing offers consistent and repeatable test outcomes, enhancing the reliability and accuracy of the test outcomes.
4. **Cost-Efficient:** Although there is an initial cost involved in setting up automated testing, in the long run, it is highly cost-efficient.

5. **Reusability:** Automated test cases can be reused across multiple releases, guaranteeing that they are consistently and frequently tested over time.